

KUSHAGRA PETWAL

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SKILLS

Languages: Embedded C, Java, FreeRtos
Embedded Platforms: STM32,ESP32
Protocols: UART, SPI, I2C, MQTT, CAN
GUI / Tools: TouchGFX, LVGL, Fusion 360, PrusaSlicer
Other: Buildroot, ARM Cortex-M/A, GNU Make, PID Control, Kalman Filter

EDUCATION

G. B. Pant University of Agriculture and Technology 2027
B.Tech 3rd Year, Electronics and Communication Engineering Pantnagar, Uttarakhand
• Member of The Robotics Club and Vimarsh: The Oratory Club

EXPERIENCE

Embedded Intern July 2025 – September 2025
Sunfox Dehradun, Uttarakhand
• Developed AFE4490 breakout-board drivers and algorithms for SpO2 and heart-rate monitoring.
• Applied FIR high-pass filters to remove baseline drift from the SpO2 signal.
• Built a custom USB protocol analyzer as a cost-effective alternative to commercial tools.
• Enforced a Kalman filter for MPU6050 and analyzed the effect of motion on SpO2 signals using the Discovery Board accelerometer.

PROJECTS

Drone Firmware 2025
• Built an ESP32-based flight controller using the Arduino framework.
• Enforced a real-time PID controller with an MPU6050 IMU over I2C.
• Executed a Kalman filter for improved motion estimation.

Custom Embedded Linux for PocketBeagle 2 | Buildroot, Embedded Linux, OP-TEE, ARM 2026
• Built a custom Embedded Linux distribution for PocketBeagle 2 using Buildroot.
• Configured the bootloader, Linux kernel, root filesystem, and cross-compilation toolchain for an ARM-based platform.
• Integrated OP-TEE and configured device tree and peripheral drivers for secure embedded execution.

Battle Bot | Fusion 360, PrusaSlicer, Pico SDK 2026
• Designed and developed a 3D-printed battle bot using Fusion 360 and fabricated components using the Prusa MK4 3D printer.
• Developed firmware using the Pico SDK, decoded FlySky transmitter inputs, and enforced watchdog-based fail-safe mechanisms for reliable operation.

LEADERSHIP

President
The Robotics Club, Pantnagar
• Organized a bot-building workshop for 100+ first-year students.
• Mentored and supervised bot-building competitions.

AWARDS

2nd Runner-Up, Battle Bot Competition 2026
Secured 3rd place in the Beetleweight category at the Robo War competition organized by IIT BHU.

Runner-Up, Maze Solving Competition 2024
Secured 2nd place in the maze-solving event organized by The Robotics Club, Pantnagar during first year.